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Improving Smart Home Bug Detection Using Random Forest Algorithm Comparing With Neural Network Algorithm

INTRODUCTION

- The aim of this project is to improving smart home bug detection using random forest algorithm comparing with neural network algorithm.
- The incorporation of smart home technology has transformed our domestic environments, providing unparalleled convenience, comfort, and energy efficiency. From controlling lights and thermostats remotely to automating daily tasks, smart homes offer unprecedented levels of control and connectivity.
- As smart home systems grow more intricate, they encounter a corresponding rise in bugs and anomalies. These complexities arise from the integration of various devices, protocols, and interfaces, leading to potential points of failure and inefficiencies. Ensuring the reliability and functionality of these systems becomes paramount to their widespread adoption and seamless operation.
- To address the challenges posed by bugs and anomalies, advanced machine learning algorithms such as the Random Forest and Neural Network algorithms have been explored.

Fig 1:RF bug detector anti spy camera detector

MATERIALS AND METHODS

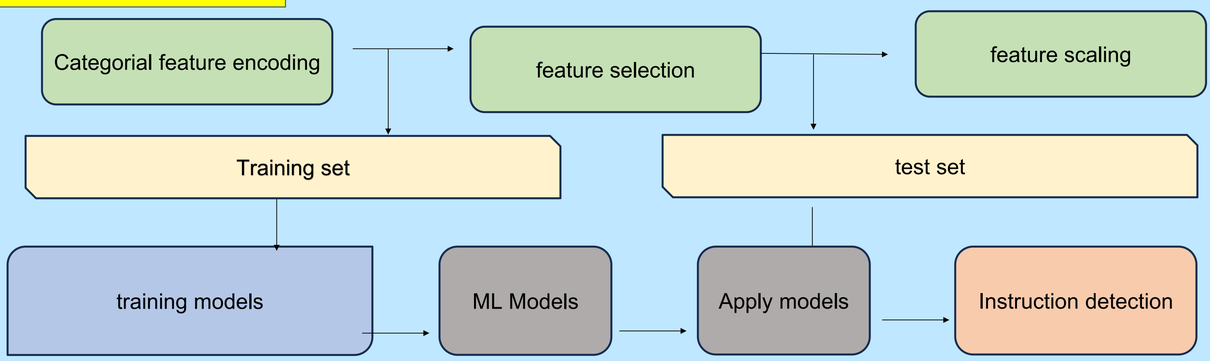

Fig 2 : smart home bug detection

RESULTS

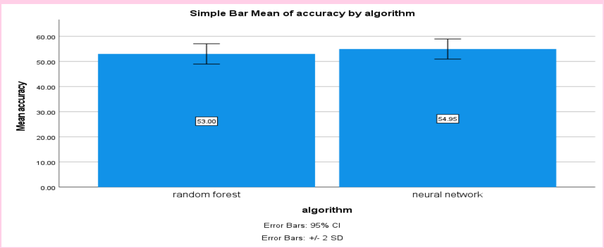

Fig 3: Random forest vs neural network

	Algorithm	N	Mean	Std.Deviation	Std.Error mean
Accuracy	Random forest	20	53.0000	2.01964	.45161
	Neural network	20	54.9500	1.99464	.44601

Table 1 : GROUP STATISTICS

The above given figure , random forest algorithm is compared with neural network algorithm, neural network has height accuracy 54.95% when compared to other algorithm.

DISCUSSION AND CONCLUSION

- The study compared the performance of two machine learning algorithms, random forest, and neural network, in predicting improving home bug detection accuracy.
- Random forest algorithm exhibited an accuracy of 53%, while the neural network algorithm had a higher accuracy of 54.95%.
- Statistical analysis revealed a significance value of $p=0.004$ ($p<0.05$), indicating that the neural network algorithm significantly outperformed the random forest algorithm .
- Mean values obtained from the random forest algorithm were consistent at 53%, whereas those from the neural network algorithm were 54.95%.
- The comprehensive evaluation focused on accuracy as a performance parameter, highlighting the neural network's superior performance with an average accuracy of 54.95% compared to random forest's 53%.

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